Appendix A
Some Laplace Transform Pairs

Laplace Transforms make it easy to represent difficult dynamic systems. A mathematical expression F(s) in the frequency domain represents a function in the time domain, a transfer function F(t) or a time function f(t). A transfer function represents the properties (or the behavior) of a mathematical block (or calculation). A time function represents a value (or signal) over time.

$F(s)$		Block Type
1		Gain Block (Gain=1)
$\frac{1}{sT}$		Integral Block
$\frac{1}{s+a}$	$T_a = 1/a$	First Order Lag
$\frac{1}{(s+a)(s+b)}$	$T_a = 1/a$ $T_b = 1/b$	Second Order Lag
$\frac{\omega}{s^2+\omega^2}$		Sine Wave (2 Interators)
$\frac{1}{s^2+2\delta\omega_n s+\omega_n^2}$		Second Order System
sT		Derivative Block
T is the Time Constant in formulas $F(s)$		

Figure A.1
Some Laplace Transform Pairs Useful for Transfer Function Analysis Fls

$F(s)$	$f(t)$
1	$Unit\ Impulse$
$\frac{1}{s}$	$Unit\ Step$
$\frac{1}{s^2}$	$Unit\ \ Ramp$
$\frac{1}{s+\alpha}$	e^{-at}
$\frac{1}{s(s+\alpha)}$	$\frac{1}{\alpha}\ (1-e^{-at})$
$\frac{1}{(s+\alpha)(s+b)}$	$\frac{1}{b-\alpha}\ (e^{-at}-e^{-bt})$
$\frac{\omega}{s^2+\omega^2}$	$\sin \omega t$
$\frac{1}{s^2+2\delta\omega_n s+\omega_n^2}$	$\frac{1}{\omega_\delta}\ e^{-\delta\omega_n t}\ \sin \omega_\delta t$ $\omega_\delta \equiv \omega_n \sqrt{1-\delta^2}$
$\frac{1}{s\ (s^2+2\delta\omega_n s+\omega_n^2)}$	$\frac{1}{\omega_\delta^2}-\frac{1}{\omega_n\omega_\delta}\ e^{-\delta\omega_n t}\ \sin(\omega_\delta t+\phi)$ $\omega_\delta \equiv \omega_n \sqrt{1-\delta^2}$ $\phi \equiv \cos^{-1}\delta$

Table A.1
Some Laplace Transform Pairs Useful for Time Function Analysis f(t)

Figure A.1 and Table A.2 show some Laplace Transform Pairs useful for control systems analysis. The output signal $f(s)_{output}$ of a block is calculated as follows:

$$f(s)_{output} = F(s)_{block} * f(s)_{input}$$

An explanation of Laplace Transform Theorems is beyond the scope of this publication and not intended [1]). Two examples will be given in Figures A.2 and A.3.

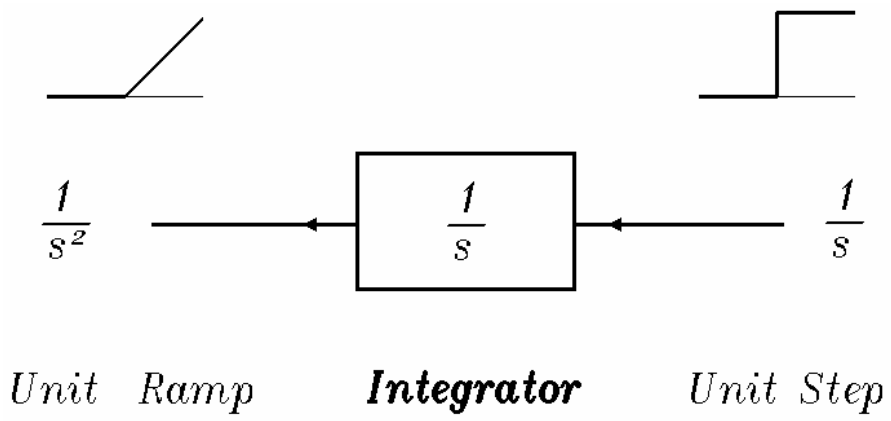

Figure A.2
Integral Block with Step Input

The Integral Block and its input, a Step Function, is a good example to show that the same function ***1/s*** in the frequency domain may represent an ***Integral Calculation*** (Block or Transfer Function) or a ***Step Function*** (Input Signal).

[1] For further studies read *"Feedback and Control Systems"* by DiStefano III, Stubberud and Williams, published by McGraw-Hill Book Company as part of Schaum's Outline Series)

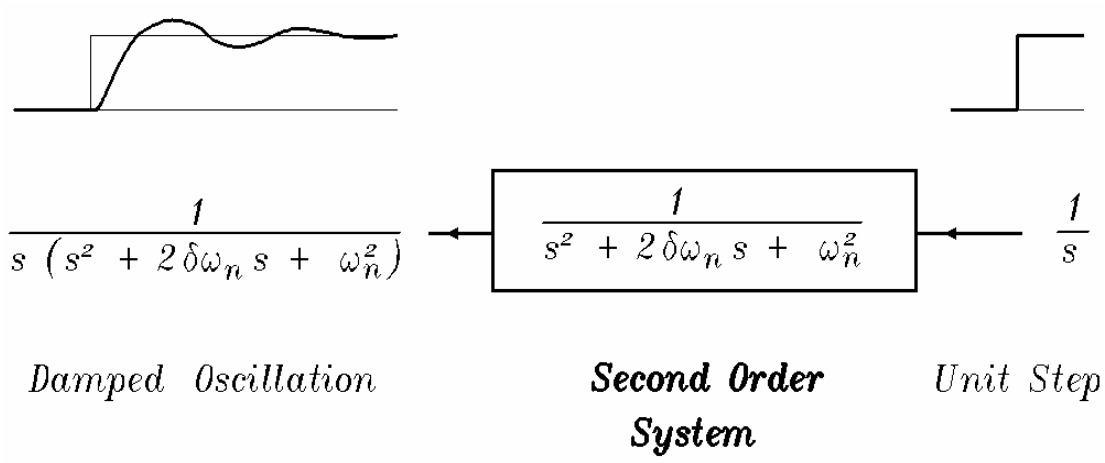

Figure A.3
Second Order System with Step Input

A Second Order System is a close representation of the behavior of many industrial processes.